

tests/lib/evidence_selftest.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** Active §1.1 paired-mutation TAP self-test for tests/lib/evidence.sh. Hermetic — runs anywhere a shell exists (bats NOT required).

Companion (§11.4.18) to the in-source documentation block at the top of the script.

Overview / Purpose

Proves every evidence.sh parser is CORRECT against captured fixtures AND provably FAILS on its negative fixture — the §1.1 paired-mutation discipline applied to the evidence harness itself: a helper that cannot catch its own negation is a bluff gate. Both polarities of every helper are asserted, including the §11.4.68 fail-open guards on assert_egress_ip (host IP unknown / empty / garbage / sentinel → OPERATOR-BLOCKED rc 2, never a fake-PASS) and the proxy_conn_verdict crashed-proxy case (a positive DIRECT signal out-ranks the port probe → FAIL, never a SKIP fail-open).

Usage

```
bash tests/lib/evidence_selftest.sh
```

No arguments.

Inputs

- None on the command line.
- Sources the library under test (tests/lib/evidence.sh) and consumes the captured fixtures under tests/lib/fixtures/ (wg snapshots, Squid access/503 logs, tcpdump / /proc/net/dev captures, egress IP files). Drives the library via its EVIDENCE_* unit-test seams — no network.

Outputs

- TAP on stdout — one ok/not ok line per case, a 1..N plan, a # tests=... passed=... failed=... line, and a # RESULT: line.
- A copy of the TAP stream written to qa-results/evidence-harness/<run-id>/selftest.tap (run-id = UTC YYYYMMDDTHHMMSSZ); the artefact path is echoed to stderr.
- Exit code: 0 iff ALL assertions pass (zero not ok); 1 if any fail.

Side-effects

- Creates qa-results/evidence-harness/<run-id>/ and writes selftest.tap there (gitignored evidence dir).
- Exports and unsets EVIDENCE_* seam variables during the run.
- No network, no live proxy, no containers — every probe is fed a fixture via a seam.

Dependencies

bash (body is POSIX-clean, parses under sh -n), awk, grep, tr, mkdir, tee. bats is NOT required.

Edge cases

- **tee subshell** → the failure count is re-derived from the artefact (grep -q '^not ok ') so the exit code is authoritative even though \$FAILS is not visible outside the tee'd pipeline.
- Asserts the hidden §15 bluff: egress==host while the host reports unknown / 0.0.0.0 must be OPERATOR-BLOCKED, never a fake-PASS.
- procdev_field value-equality assertions pin exact tx/rx packet counters from the fixture snapshot.
- proxy_conn_verdict asserts both false-FAIL (§11.4.1) and fail-open (§11.4.68) are impossible, plus _code_in whole-token matching (20 ≠ 200).

Related scripts

- tests/lib/evidence.sh — the library under test.
- tests/dynamic/lib/analyzer_common.sh — the dynamic-mode base that sources evidence.sh read-only.
- tests/regression/assert_egress_ip_host_unknown_test.sh — the F7 fail-open regression guard exercised in isolation.
- Constitution §11.4 / §11.4.68 / §11.4.69 / §11.4.107 / §1.1; design §13/§14.

Last verified

2026-07-01 — documented against the script source; body parses under `sh -n`. Runs hermetically (fixtures + EVIDENCE_* seams, no network) and emits its TAP artefact under `qa-results/evidence-harness/`.